

Dr. Ayan Sadhukhan
Course:- Introduction to Bioengineering

Evolution, The Themes of Biology, and Scientific Inquiry

1. Biology & Engineering Approaches

I) Observations; deductive, inductive and abductive reasoning; hypothesis testing

[Characteristics of Life](#)

[The Scientific Method](#)

[Scientific Method](#)

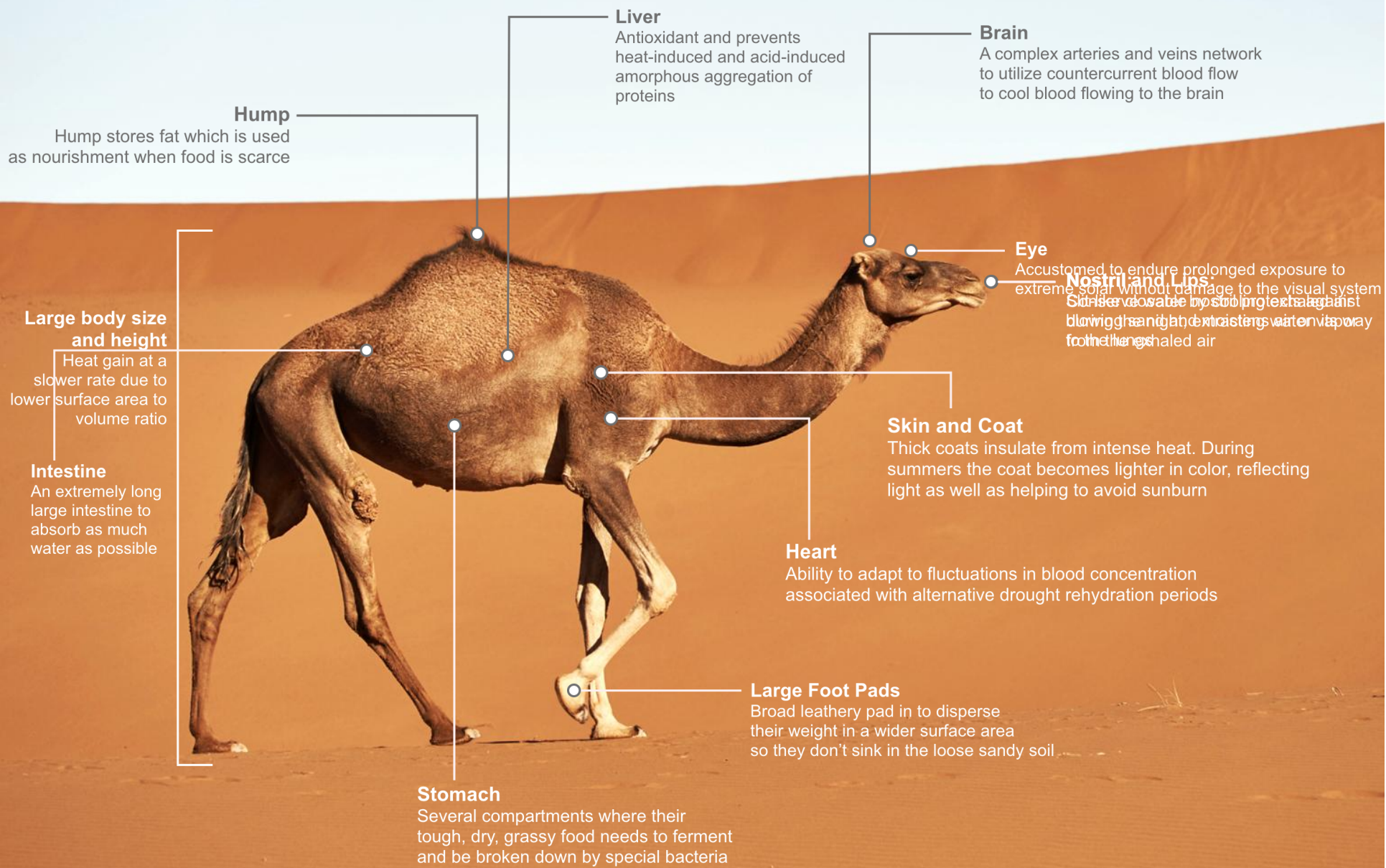
[Inductive Reasoning](#)

[Deductive Reasoning](#)

[The Scientific method -](#)

<https://youtu.be/xOLcZMw0hd4?si=FXRjBJgd8jXkOVbC>

DESIGN NEEDS & INSPIRATION



Arid Environmental Stresses

- Heat
- 'Water Scarcity'
- Food Scarcity
- Sand

Design & Applications

Tire designed with two lobes of braided steel to be able to easily trek over hills of fine-grained, abrasive lunar dust
Researchers in the Sahara Desert are using Camel's nose technique to remove salt from saltwater using cool ground water to evaporate



Silent Spring



Human Lice



Co-Evolution



Head louse, *Pediculus capitis*
- lives and feeds exclusively
on the human scalp



Body louse, *P. humanus*
- feeds on the body, lives
and lays eggs in clothing

https://www.youtube.com/watch?v=zMHatXh288k&ab_channel=ScienceDream